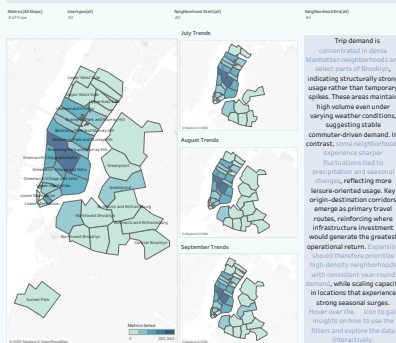


The chart illustrates the change in trip demand over time, categorized by Domestic and International travel. The Y-axis represents the demand level, ranging from 0.00 to 0.05. The X-axis shows the timeline from December 2019 to December 2020. Domestic demand (dark blue) shows a significant peak in late 2019, followed by a decline and then a second peak in late 2020. International demand (light green) remains relatively low throughout the period.

Month	Domestic	International
December 2019	0.015	0.005
March 2020	0.010	0.005
June 2020	0.040	0.010
September 2020	0.045	0.010
December 2020	0.025	0.010
March 2021	0.015	0.005
June 2021	0.040	0.010
September 2021	0.045	0.010
December 2021	0.035	0.010

Casual riders increase sharply in warmer months, showing that weather plays a big role in short-term spikes. The year-over-year growth during peak seasons also signals rising overall demand. Together, this suggests expansion should focus on areas with strong subscriber activity for stability, while also preparing for higher bike and dock capacity in locations that see large seasonal increases from casual riders.



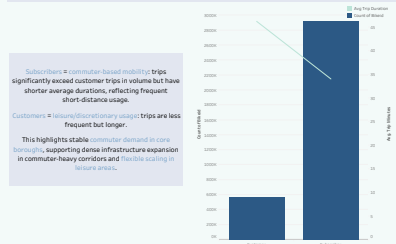
Total Trips By Starting Station

[illegible]

The Top 10 neighborhoods by trip volume show a clear concentration of activity in dense Manhattan areas such as Chelsea, Lower East Side, and Greenwich Village, with Northwest Brooklyn also emerging as a strong contributor. The consistency between high starting and ending trip volumes suggests these areas function as both major origins and destinations, indicating established commuter and leisure corridors. The high trip minutes further reinforce sustained usage rather than short, isolated rides, highlighting structurally strong demand in these zones.

Segment	Revenue Split	Bookings	Manufacture	Revenue Split	Quantity	Grand Total
Customer	Bookings	75,822	76,950		26,200	179,894
	Manufacture		490,475		40,200	490,475
	Quantity	0.000	0.000		0.000	
Subsidiary	Bookings	239,750	239,020		124,170	400,728
	Manufacture	0.000,000	0.000,000		0.000,000	0.000,000
	Quantity	0.000,000	0.000,000		0.000,000	0.000,000
Grand Total		2,000,700	0.000,000		0.000,000	0.000,000

Initial neighborhood-level analysis showed that the highest trip volumes were concentrated in Manhattan. A deeper review of borough start and end patterns confirmed that most trips both originate and terminate within Manhattan, with subscriber rides driving the majority of this activity. Intra-borough travel significantly outweighs cross-borough movement, indicating that riders primarily use the system for short, localized commuting rather than long-distance travel. Expansion should therefore prioritize increasing station density and dock capacity within high-demand Manhattan corridors, while selectively scaling infrastructure in strong-performing Brooklyn areas and targeted zones in Queens.



Trip demand is strongest and cost constraints in dense Manhattan neighborhood, driven primarily by commuters using the system for short, frequent commuter trips. **Causal riders contribute** in seasonal and, especially, in warmer months, to the system's overall capacity. **Based on the concentration of demand and consistent bidirectional flow in these neighborhoods**, expansion efforts should prioritize high-volume Manhattan corridors while selectively scaling infrastructure in strong performing Brooklyn areas. **Investment should** focus on increasing dock capacity and station density in locations that demonstrate both high origin and destination activity, ensuring operational efficiency and return on expansion. Overall, this approach supports stable year-round service in commuter-heavy areas while maintaining flexibility to meet seasonal leisure demand.

The analysis combines multiple publicly available datasets: [NYC Citi Bike Trip Data](#) for trip history, [ZIP Code Boundaries](#) for neighborhood and borough mapping, and [NOAA GSDO Weather Data](#) for daily temperature, wind speed, and precipitation. These datasets were integrated using SQL and joined via geospatial points and dates to create a unified analytical dataset. Trip data from 2014–2015 was shifted 5 years forward to align with more recent dates, and summer months were analyzed separately to examine seasonal effects on trip demand.